higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N4

2 DECEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 A pentaprism is a prism with a five-sided cross section and two silvered surfaces✓ giving a constant deviation of all rays of light through 90° and used chiefly in the viewfinders of single-lens reflex cameras.✓

1.2 Direction is the angle measured clockwise✓ between zero direction of the coordinates system and the line in question.✓

1.3 Chain surveying occurs when only linear measurements✓ are made to locate points using a tape.✓

1.4 Coordinates are values (in meters) of a point on the ground represented on a map✓ or plan and it consists of an ordinate y value and an abscissa – the x value.✓

1.5 The prime meridian is the zero meridian✓ from which all longitudes are measured.✓

(5 × 2) **[10]**

QUESTION 2

2.1
- Horizontal distance
- Vertical distance
- Horizontal angle
- Vertical angle (4)

2.2
- Fieldwork
- Office work
- Care and adjustment of instruments (3)

2.3
- Accuracy
- Integrity
- Legibility
- Arrangement
- Clarity (5)

[12]

QUESTION 3

3.1 P = Applied tension in N
PO = Standard tension for particular tape in N
1 m = Length of the tape in mm
A = Cross-sectional area of the tape in mm
E = Module of elasticity of the steel (5)

3.2 $$\text{Area} = \frac{1}{2} ab \sin c \text{ v}✓$$

$$15\,600 \text{ m}^2 = \frac{1}{2} \times 250 \times b \sin 30° \text{ v}✓$$

$$31\,200 \text{ m}^2 = 250 \times b \times \sin 30° \text{ v}✓$$

$$b = AC = \frac{31\,200}{250 \times \sin 30° \text{ v}✓}$$

$$= 249{,}60 \text{ m}✓ \quad (5)$$

3.3 $$\frac{X}{100^g} = \frac{45°}{90°}✓$$

$$X = \frac{4\,500}{90}✓$$

$$X = 50^g✓✓ \quad (4)$$

3.4 Area = L × B✓
= 160 m × 80 m✓
= 12 800 m^2✓✓ (4)
[18]

QUESTION 4

4.1 The instrument is held against a ranging rod and aimed at a distant ranging rod in such a way that the line of sight is parallel to the ground.✓ The bubble is then turned by a large milled-head knob✓ until it appears in the mirror.✓ When the bubble is centered against the cross-wire, the index arm will be pointing to the vertical angle and the reading is noted.✓ A vernier scale and magnifying glass may be provided for finer reading.✓ (5)

4.2 The distance is measured by calculating the time difference between a transmitted pulse✓ and a return pulse or the phase difference between a transmitted✓and a reflected beam of radiation.✓ (3)

4.3
- The tape should be drawn out in a straight line in the direction in which it is coiled.
- Ensure that no kinks are present.
- Wipe the tape after use with an oily cloth to prevent rust.
- Do not allow a vehicle to run over the tape.
- Avoid stepping on the tape while it is lying on the ground.
- Never leave the tape lying on the ground for longer than necessary.
- Do not apply too much tension. (7)

[15]

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QUESTION 5

5.1 $$CD = MD \times \frac{ALT}{GLT}$$

$$\frac{50+0,03}{50}$$

$100 = MD \times$ ✓

$100 = MD \times (1,0006)$✓

$$\frac{100}{1,0006} = MD$$✓

$MD = 99,94$ m✓ (4)

5.2 Ct = L × e × (Tm – Ts)
= 60 × 1,13 × 10^{-5} × (15 – 20)✓
= -0,003 m✓
CD = 60 – 0,003✓
= 59,997 m✓ (4)

5.3 Cs = L (1 – cosΘ)
= 95,45(1 – cos 7° 40' 20")✓
= 0,854 m✓
CD = 95,45 – 0,854✓
= 94,596 m✓ (4)

5.4 $$Ch = \frac{L \times H}{R}$$

$$= \frac{300 \times 1\,450}{6\,373\,000}$$ ✓

$= 0,068$✓

$CD = 300 + 0,068$✓

$= 300,068$ m✓ (4)
[16]

QUESTION 6

6.1
- Traveller or boning rod
- Change plate
- Staff bubble
- Levelling staff
- Tripod stand (5)

6.2 Parallax is, when looking through a telescope,✓ the apparent movement of the cross hairs relative to the object onto which the instrument is being sighted.✓ To remove parallax, observe the sky though the telescope✓ and adjust the eyepiece out or in until the cross hairs appear in sharp focus. The eyepiece is now focused in the reticule.✓ Sight a distant object and focus the telescope until the image of the object is clearly defined.✓ The eyepiece is now clearly focused on the cross hairs and the image✓ and if the image of the object and cross hairs do not move relative to each other, the parallax has been eliminated.✓ (7)

[12]

QUESTION 7

7.1
7.1.1 Perpendicular offset
7.1.2 Length and direction
7.1.3 Ties
7.1.4 Forward intersection

(4 × 1) (4)

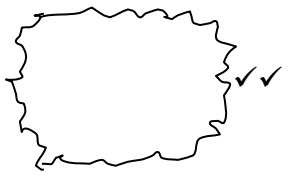
7.2
7.2.1 2,085
7.2.2 2,030
7.2.3 1,980

(3 × 1) (3)

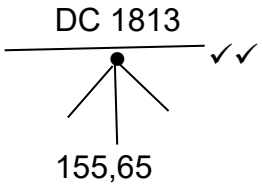
[7]

QUESTION 8

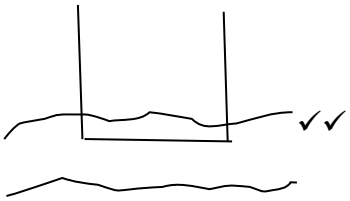
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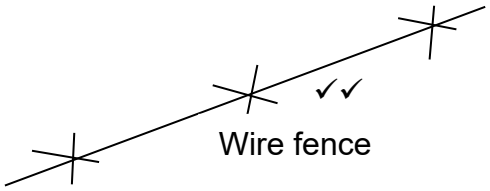
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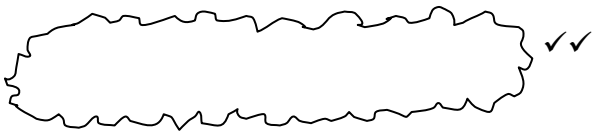
8.3



8.4



8.5



(5 × 2) **[10]**

TOTAL: 100

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